

Fall 1996

Problem 1 (Fa96). Let M be the set of real valued continuous functions f on $[0, 1]$ such that f' is continuous on $[0, 1]$, with the norm

$$\|f\| = \sup_{0 \leq x \leq 1} |f(x)| + \sup_{0 \leq x \leq 1} |f'(x)|$$

Which subsets of M are compact?

Problem 2 (Su85, Fa96). A function $f : [0, 1] \rightarrow \mathbb{R}$ is said to be *upper semicontinuous* if given $x \in [0, 1]$ and $\varepsilon > 0$, there exists a $\delta > 0$ such that if $|y - x| < \delta$, then $f(y) < f(x) + \varepsilon$. Prove that an upper semicontinuous function f on $[0, 1]$ is bounded above and attains its maximum value at some point $p \in [0, 1]$.

Problem 3 (Fa79, Fa80, Sp85, Su85, Fa96, Fa98, Sp99, Sp26). Prove that

$$\int_0^\infty \frac{x^{\alpha-1}}{1+x} dx = \frac{\pi}{\sin \pi \alpha}$$

What restrictions must be placed on α ?

Problem 4 (Fa96, Fa16). Does there exist a function f , analytic in the punctured plane $\mathbb{C} \setminus \{0\}$, such that

$$|f(z)| \geq \frac{1}{\sqrt{|z|}}$$

for all nonzero z ?

Problem 5 (Su79, Fa96). Prove that a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ has

1. a one-dimensional invariant subspace, and
2. a two-dimensional invariant subspace.

Problem 6 (Su80, Fa96). Let A and B be real 2×2 matrices such that $A^2 = B^2 = I$ and $AB + BA = 0$. Prove there exists a real nonsingular matrix T with

$$TAT^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad TBT^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Problem 7 (Fa96). Suppose p is a prime. Show that every element of $\text{GL}_2(\mathbb{F}_p)$ has order dividing either $p^2 - 1$ or $p(p - 1)$.

Problem 8 (Fa96). Show the denominator of $\binom{1/2}{n}$ is a power of 2 for all integers n .

Problem 9 (Fa84, Fa96, Fa10, Fa18). Let gcd abbreviate greatest common divisor and lcm abbreviate least common multiple. For three nonzero integers a, b, c , show that

$$\gcd\{a, \text{lcm}\{b, c\}\} = \text{lcm}\{\gcd\{a, b\}, \gcd\{a, c\}\}.$$

Problem 10 (Fa96). If f is a C^2 function on an open interval, prove that

$$\lim_{h \rightarrow 0} \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} = f''(x)$$

Problem 11 (Fa82, Fa96). Let f be a real valued continuous nonnegative function on $[0, 1]$ such that

$$f(t)^2 \leq 1 + 2 \int_0^t f(s) \, ds$$

for $t \in [0, 1]$. Show that $f(t) \leq 1 + t$ for $t \in [0, 1]$.

Problem 12 (Su80, Fa83, Sp84, Fa87, Fa96, Fa07). How many zeros does the complex polynomial

$$3z^9 + 8z^6 + z^5 + 2z^3 + 1$$

have in the annulus $1 < |z| < 2$?

Problem 13 (Fa77, Fa96, Sp19). Let $\mathbb{C}^3$ denote the set of ordered triples of complex numbers. Define a map $F : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ by

$$F(u, v, w) = (u + v + w, uv + vw + wu, uvw)$$

Prove that F is onto but not one-to-one.

Problem 14 (Su83, Fa96). Let f be analytic on and inside the unit circle $C = \{z \mid |z| = 1\}$. Let L be the length of the image of C under f . Show that $L \geq 2\pi|f'(0)|$.

Problem 15 (Fa96, Sp16). Is there a real 2×2 matrix A such that, for $\varepsilon > 0$

$$A^{20} = \begin{pmatrix} -1 & 0 \\ 0 & -1 - \varepsilon \end{pmatrix}?$$

Exhibit such an A or prove there is none.

Problem 16 (Fa96). Let $A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$. Show that every real matrix B such that $AB = BA$ has the form $B = aI + bA + cA^2$ for some real numbers a , b , and c .

Problem 17 (Fa96). Let $\mathfrak{J} \subset \mathbb{Z}[x]$ be the ideal generated by 13 and $x - 4$. Find an integer m such that $0 \leq m \leq 12$ and

$$(x^{26} + x + 1)^{73} - m \in \mathfrak{J}$$

Problem 18 (Su80, Fa96, Sp11). Prove that every finite group is isomorphic to

1. A subgroup of a group of permutations of n objects;
2. A subgroup of the group of even permutations of n objects.